

• INTERACTIVE DEMO • ILLUSTRATIVE APPLICANT • SYNTHETIC DATA

Some of your best borrowers don't have a credit file to prove it.

This is a short, open-source demonstration of how traditional credit information can be supplemented with everyday financial signals — rent, utilities, cash flow — to produce an assessment a loan officer can actually read, question, and check for fairness. Not a product pitch. A working example, walked through with one fictional applicant.

[Walk through the example ↓](#)[View the code on GitHub](#)

THE IDEA IN THREE PARTS

01

Look beyond the bureau file

With an applicant's consent, weigh rent, utility, and cash-flow patterns alongside — not instead of — traditional credit data.

02

Show the reasoning

Every assessment comes back with plain-language reasons, so a human can see why — not just a score.

03

Check for fairness

Built-in metrics flag when approval rates diverge sharply across groups, so it's monitored rather than assumed.

WALKTHROUGH

Meet the applicant

Six steps, one fictional small business. Steps 3 to 6 use the platform's actual data fields, model, and logic — the assessment, the reasons, and the fairness numbers are live output. Steps 1 and 2 are the situation a loan officer walks in with; the platform never sees them.

Maria's Bakery — working-capital application

FICTIONAL EXAMPLE

1. Applicant profile

Maria has run her bakery for three years. Revenue is steady but seasonal, and — like many small business owners — most of her financial life runs through her business checking account and her landlord, not a credit card.

Business	Maria's Bakery
Structure	Sole proprietor, owner-operated
Time in business	3 years

Requesting	\$35,000 working capital
Purpose	Second oven, seasonal demand

What the platform actually scores here is Maria the person, not the bakery as an entity: her rent, her utilities, and the cash flow through her account, used as a proxy for the business. That is how CDFIs and micro-business lenders already underwrite owner-operated borrowers, and it is what this feature set supports today — there is no business-specific contract yet. The amount and purpose above are lender context; the model has no loan-size, term, or debt-to-income input, so this profile would score identically for \$5,000 or \$350,000.

2. Traditional credit file

On paper, there isn't much to go on. A thin file like this is common — by some estimates, tens of millions of U.S. adults are considered credit invisible or unscorable using conventional methods alone. It doesn't mean someone is risky. It means the bureau file has little to say.

Credit file status	Thin file
Open trade lines	1 (secured card, opened 4 months ago)
Loan history on file	None
Conventional bureau score	Not yet scoreable

The widely used bureau models need at least one account open for six months or more, so a card opened four months ago does not yet produce a conventional score. Note what this step is and isn't: it is the situation a loan officer faces, not an input to the platform. The model never receives any of the four fields above — it reads only the nine alternative-data fields in the next step.

3. Additional financial signals

With Maria's consent, the platform can also look at day-to-day financial behavior — the kind of information her landlord and utility provider already have, but a credit bureau doesn't.

PAYMENT HISTORY

Rent, on time (12 mo)	100%

Utilities, on time (12 mo)

96%

Electric and gas, 23 of 24 payments on time.

CASH FLOW (6 MO)

Avg. monthly income

\$5,400

Avg. daily balance

\$2,150

Volatility index

0.285

ACCOUNT RISK EVENTS (12 MO)

Overdrafts

0

Non-sufficient-funds events

0

STABILITY

Months at current address

24

Months at current job

36

Self-employed — the bakery is the job.

These nine values are the complete set the model receives — nothing else from the previous steps reaches it. The rent figure is Maria's home lease, which is the payment record a landlord can verify; the volatility index is a normalised 0–1.5 measure where the reference profile sits at 0.24, so 0.285 reads as mildly seasonal.

4. Credit assessment

The assessment isn't a yes/no gate — it's a starting point an underwriter can read, question, and combine with their own judgment and policy.

53.7
/100

Above the review threshold, with one factor flagged

Read this as a probability of on-time repayment on a 0–100 scale, not as a credit score. For calibration: an applicant

sitting exactly on the platform's published reference profile scores **51.8**, and the strongest inputs the contract allows reach **81.1**. Maria clears the threshold on the strength of her verified income, with one factor flagged for a human to weigh — see the next step.

This is the actual output of the platform's baseline model for this scenario — live-scored against the real API, not staged. That model is trained on synthetic demonstration data only and its baseline ROC-AUC is modest, which is why the scores above cluster near the middle of the range. Treat the number as illustrative of the workflow, not as a production-grade score, and not as a real underwriting decision.

5. Key factors behind this assessment

Instead of a black-box score, the system returns plain-language reasons — the same kind of information adverse-action notices are meant to convey under fair lending law.

✓ WHAT HELPED

- ✓ Verified monthly income well above the reference profile — by far the largest positive
- ✓ Three years running the bakery, entered as job tenure
- ✓ Utility payments ahead of the reference on-time rate
- ✓ Rent paid on time every month, though it moves this baseline only slightly

⚠ WORTH WATCHING

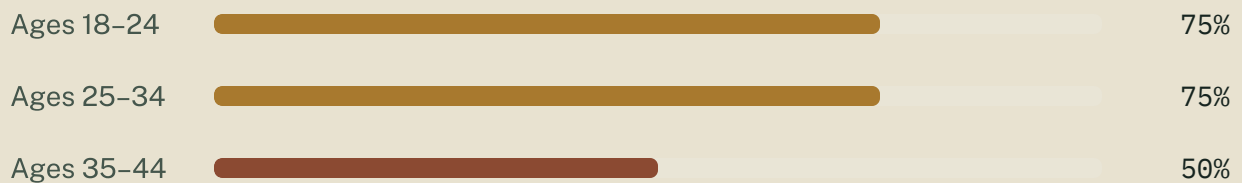
- ⚠ Seasonal cash-flow swings — the only reason code the model raised

Neutral: no overdrafts and no NSF events in the past 12 months, and two years at the same address — each sits exactly on the reference profile, so neither helps nor hurts.

An honest reading of the explainability panel further down: on this baseline, income and cash-flow volatility do nearly all the work, and rent history moves the score far less than the premise of this project implies. Dropping Maria's rent from 100% to 50% on time costs about one point and does not change the decision. That is a weakness of a model trained on synthetic data where almost every simulated applicant already pays rent on time — and it is exactly the kind of thing the explanation layer exists to surface. Making rent history carry its weight is open modelling work, listed in the project's own risk register rather than hidden behind the score.

6. Fairness check

This isn't Maria-specific — it's a portfolio-level check a lender runs on demand against a batch of decisions. Below: selection rate by age group, computed live by the fairness endpoint from the console's built-in monitoring batch.



Gap between highest and lowest group: **25 percentage points** — large enough that the metrics came back flagged for review rather than stable. That's the check working as intended: surfacing a signal for a human to look at, not quietly folding it into an approval rate.

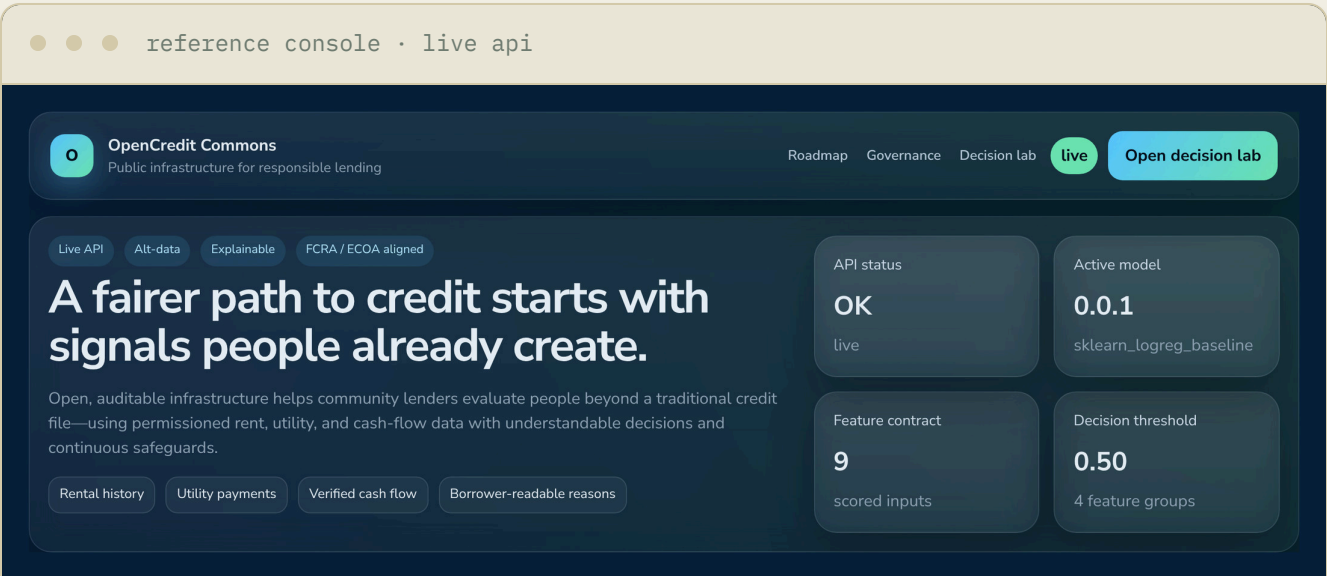
Be precise about what this does and doesn't show. The monitoring batch is a fixture: its decisions and outcomes are pre-set in the console so the metrics have something deterministic to measure, and the model did not produce them. The 25-point gap is a property of that fixture, not evidence about this model. When the model does score a real cohort — the 24-application portfolio pass shown further down — the selection rate comes back identical across every age group. Measuring a live model this way needs a lender's own decisions and observed outcomes, which is what the outcome feedback loop is for.

PROOF, NOT MOCKUPS

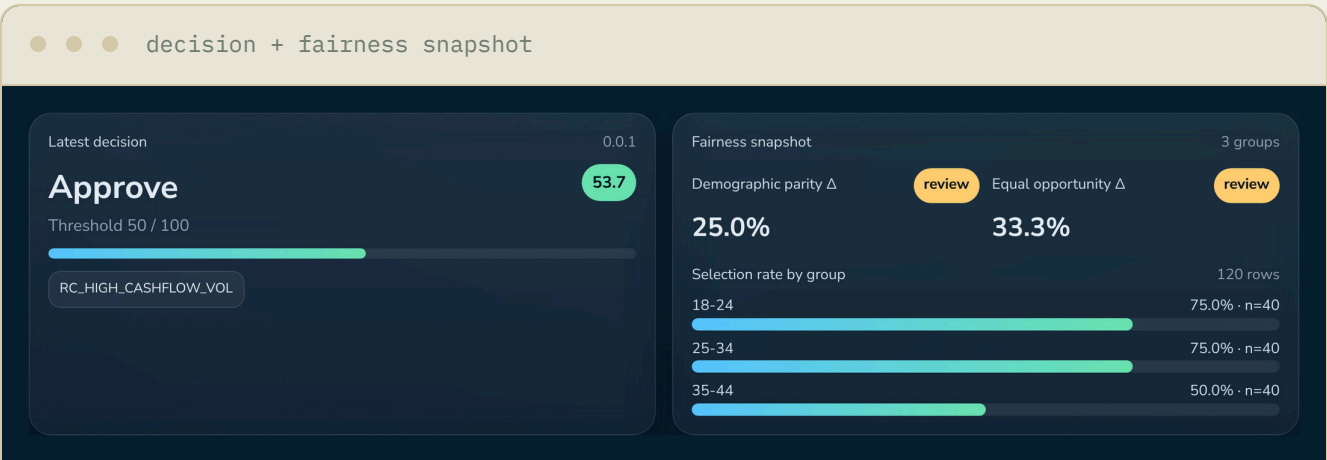
This is what's actually running

Screenshots below, not illustrations — the open-source reference console, scoring the Maria's Bakery scenario above through the live API and the real

trained baseline model. One continuous session, captured end to end: the applicant goes in, a decision comes out, and every explanation, fairness check, and audit record on this page came from that run.



The reference console from this repository, running against the live scoring service — API status, the active model version, and the feature contract it exposes.



The live result: 53.7 / 100, approve, with RC_HIGH_CASHFLOW_VOL raised — and beside it the fairness snapshot, both metrics flagged for review rather than marked stable.

BORROWER TRANSPARENCY PREVIEW

A decision people can understand—and question.

This is the plain-language layer generated from the same governed decision record used by the lender.

Application outcome

Eligible to proceed

Reference 572ff3e3-f3f0-4773-b5d3-962613f3c0c6

[View your rights](#)

WHAT MOST AFFECTED THIS REVIEW

1 FACTORS

1 Recent cash flow varied more than expected

Month-to-month deposits and expenses showed less consistency during the review period.

Providing a longer verified history may show a more stable pattern.

No protected characteristic is used to generate the score.

Human review available

The same decision translated for the applicant — the factor that mattered, and what could change it.

Explainability linear_proxy

Explain request: d98a1b5e-d15a-47b9-80ac-432502ac082f

Application: maria_bakery_demo

Latest score request: 572ff3e3-f3f0-4773-b5d3-962613f3c0c6

Log-odds movement versus the reference profile in the feature contract. Positive values push toward approval; the values sum to the gap between this applicant and a typical one.

Average monthly income (6m)	+0.079
Cashflow volatility (6m)	-0.038
Months at current job	+0.016
Utility on-time rate (12m)	+0.013
Rent on-time rate (12m)	+0.005
Average daily balance (6m)	+0.003
NSF events (12m)	+0.000
Overdraft events (12m)	+0.000
Months at current address	+0.000

Contributions measured against the reference profile the feature contract publishes, so they sum to the gap between Maria and a typical applicant. Income does most of the work; rent moves this baseline very little.

Fairness monitor

Synthetic approval/outcome batch mapped to monitoring groups

18-24 n=40
Selection rate 75.0%
Positive outcome rate: 75.0%

25-34 n=40
Selection rate 75.0%
Positive outcome rate: 75.0%

35-44 n=40
Selection rate 50.0%
Positive outcome rate: 75.0%

Use this to simulate parity checks before plugging in aggregated outcomes from a deployment environment.

[Regenerate batch](#)

[Run fairness report](#)

The batch behind that 25-point gap: a fixture with pre-set decisions, 40 rows per group, so the metrics have something deterministic to measure. The model did not produce these approvals.

Score applicant Contract-driven fields from the active model

Reset defaults

Explain only

Score applicant

Every input below is generated from the backend feature contract, so the UI stays aligned with the current model schema and threshold.

Model: sklearn_logreg_baseline

Version: 0.0.1

Threshold: 50 / 100

Schema: a991402390c9a9aa

Application identifier

Hashed in audit payloads

maria_bakery_demo

payment history 2 signals

Rent on-time rate (12m) 1.00

Share of observed rent payments made on time during the last 12 months.

1

Required higher helps

Utility on-time rate (12m) 0.96

Share of observed utility payments made on time during the last 12 months.

0.96

Required higher helps

cash flow 3 signals

Average monthly income (6m) 5400

Average verified monthly income over the last six months.

5400

Required higher helps

Cashflow volatility (6m) 0.28

Normalized volatility of recent cash flow, where higher values mean less stability.

0.285

Required lower helps

Average daily balance (6m) 2150

Average account balance over the last six months.

2150

Required higher helps

risk events 2 signals

NSF events (12m) 0

Count of non-sufficient funds events observed over the last 12 months.

0

Required lower helps

Overdraft events (12m) 0

Count of overdraft events observed over the last 12 months.

0

Required lower helps

stability 2 signals

Months at current job 36

Tenure at the current job, when available.

36

Optional higher helps

Months at current address 24

Tenure at the current address, when available.

24

Optional higher helps

Age band Optional monitoring context

35-44

Race / ethnicity Aggregated only

decline_to_state

Sex Aggregated only

decline_to_state

And the inputs behind it: Maria's Bakery entered into the console. Every field on this screen is generated from the model's own feature contract, so the form cannot drift from the schema being scored.

portfolio workbench

Portfolio workbench Generate cohort snapshots against the active model

Cohort size 24 applications

12

24

40

60

Fairness grouping used only for monitoring

age_band

Label coverage 12 positive outcomes

The sample cohort carries synthetic outcomes so the backend can compute equal opportunity deltas as part of the analysis.

Run portfolio analysis

Portfolio summary

0.0.1

Average score

28.3%

Approval rate

16.7%

Applications

24

Threshold

50/100

Top reason codes

5 returned

RC_OVERDRAFT_EVENTS

Overdraft events were observed in recent history.

20

RC_NSF_EVENTS

Non-sufficient funds events were observed in recent history.

18

RC_LOW_RENT_ON_TIME

Recent rent payment history indicates lower on-time payment consistency.

16

RC_HIGH_CASHFLOW_VOL

Cash-flow patterns show higher volatility, increasing repayment uncertainty.

16

RC_LOW_UTIL_ON_TIME

Recent utility payment history indicates lower on-time payment consistency.

6

A 24-application cohort scored in a single pass, with the reason codes that drove the outcome mix.

outcome feedback

PERFORMANCE FEEDBACK LOOP

Connect decisions to what happened next.

Verified repayment outcomes power calibration checks, fairness monitoring, and governed retraining. Direct personal identifiers are not required here.

Decision → Observed outcome → Governance review →

Approved update

Application identifier

required

maria_bakery_demo

Observation window

90-day repayment — Initial performance window

Observed result

Paid as agreed

Positive performance

Not paid as agreed

Adverse performance

Outcome recorded and governance metrics refreshed.

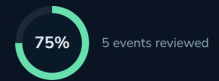
Record outcome

Observed repayment outcomes are recorded against the decision, which is what makes calibration and fairness monitoring measurable rather than assumed.

CONTINUOUS GOVERNANCE

Evidence before automation.

Every decision should be reproducible, reviewable, and attached to the controls that made it safe to issue.



Control register

Decision lineage

Monitoring signals

● Decision explanation coverage 100% ● Observed outcome coverage 100% ● Latest fairness threshold review 33% ● Decision audit capture 100%

GV-01 DATA RIGHTS DATA GOVERNANCE

READY

Consumer permission and permissible-purpose record

Consent receipt, source reference, purpose, and expiration are bound to each ingestion event.

GV-02 DATA QUALITY ML PLATFORM

READY

Alternative-data lineage and reproducibility

Dataset versions, transformations, feature schema, and training inputs are traceable end to end.

GV-03 FAIR LENDING RESPONSIBLE AI

READY

Pre-release subgroup evaluation

Selection rate, equal opportunity, calibration, and threshold sensitivity are reviewed by subgroup.

GV-04 EXPLAINABILITY COMPLIANCE

REVIEW

Specific and actionable decision reasons

Feature contributions map to stable reason codes and plain-language adverse-action narratives.

GV-05 MODEL RISK MODEL VALIDATION

PLANNED

Independent validation and stress testing

Performance, stability, fairness, and explanation consistency are tested under adverse scenarios.

GV-06 CHANGE CONTROL MODEL GOVERNANCE

REVIEW

Approval, rollback, and incident handling

Registry promotion requires named approval; prior artifacts remain available for rapid rollback.

The control register, with coverage metrics computed from the audit events this run just produced.

Audit explorer 3 events indexed

All event types ▾

maria_bakery_demo

When	Type	Request	Application	Model	Payload
8/26/2026, 1:31:13 PM	outcome	—	ref_a92750ffca8d5caeb4b89dcfb403891d4a985199f0df9f2739f2d523566c11ec	—	{ "outcome_type": "repayment_90d", "outcome_value": 1 }
8/26/2026, 1:31:06 PM	explain	d98a1b5e-d15a- 47b9-80ac- 432502ac082f	ref_a92750ffca8d5caeb4b89dcfb403891d4a985199f0df9f2739f2d523566c11ec	0.0.1	{ "contribution_coun t": 9, "reason_codes": ["RC_HIGH_CASHFLOW_ VOL"], "score": 0.5370427978848056 }
8/26/2026, 1:31:04 PM	decision	572ff3e3-f3f0- 4773-b5d3- 962613f3c0c6	ref_a92750ffca8d5caeb4b89dcfb403891d4a985199f0df9f2739f2d523566c11ec	0.0.1	{ "score": 0.5370427978848056 , "decision": "approve", "decision_threshol d": 0.5, "reason_codes": ["RC_HIGH_CASHFLOW_ VOL"], "features_hash": "1ceca67fc7481d9e8 b62f059e0917deceb8 7aa584b4c6ef57972b d84766869bd" }

The append-only record for this one application — decision, explanation, and observed outcome — with the identifier stored as a pseudonymous hash.

How a lender might use this alongside existing underwriting

The platform is built to sit next to a lender's process, not replace it.

01

Officer reviews

A loan officer sees the applicant, the assessment, and the plain-language reasons behind it.

02

Weighed against policy

That information is combined with the institution's own underwriting policy, documentation, and judgment.

03

Lender decides

The lending decision — and the compliance responsibility for it — stays with the lender, not the platform.

SCOPE

What this is — and isn't

THIS IS

- + An open-source framework for alternative-data credit assessment
- + A demonstration environment using synthetic data
- + A way to explore explainable, fairness-aware scoring approaches
- + Something a technical team could pilot, test, or extend

THIS IS NOT

- A credit bureau or a production scoring service
- An automatic approve/deny system
- A guarantee of improved lending outcomes
- Currently in use by any financial institution
- Legal or compliance advice

UNDER THE HOOD

The full implementation — feature definitions, the scoring API, fairness metrics, and audit logging — is public.

Apache-2.0 license · github.com/WangProjects/credit-evaluation-platform

View repository →

This page is a demonstration built on synthetic data and a fictional applicant. It does not represent an actual lending decision, is not a credit bureau, does not replace any institution's underwriting process, and is not currently deployed by any financial institution. Nothing here is legal or compliance advice — any real deployment should be reviewed with qualified counsel.

Inclusive Credit Infrastructure — Apache-2.0 [GitHub](#) jiaye_wang@berkeley.edu